

Sharp Universal Metric Distortion into Every Schatten Class

Candidate full solution for arXiv:2405.03517, Question 2

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Status. Candidate full solution, likely valid, pending human review. The argument answers the source question for every $1 \leq p < \infty$, including the trace-class endpoint $p = 1$.

Abstract

Escudero Gutierrez and Muguruza ask for the worst-case distortion D_{n,S_p} of an n -point metric space into the Schatten class S_p . We prove, with universal implicit constants,

$$D_{n,S_p} \asymp \max \left\{ 1, \frac{\log n}{p} \right\} \quad (1 \leq p < \infty).$$

For $p \geq 2$, a lazy-walk interpolation argument gives $\tilde{h}_{S_p}(G) \geq \delta/(14p)$ for every regular graph with normalized adjacency gap δ . The missing range $1 \leq p \leq 2$ requires a different mechanism. Translate a Schatten-valued map at a well-chosen vertex and apply the noncommutative Mazur map $M_{p,2}(x) = u|x|^{p/2}$ into S_2 . Ricard's uniform Holder and inverse Lipschitz estimates, a weighted averaging lemma, and the Hilbert Poincare inequality yield

$$\tilde{h}_{S_p}(G) \geq (\delta/K)^{1/p} \quad (1 \leq p \leq 2)$$

for a universal K . Fixed-gap expanders then give the required lower bounds, while the known ℓ_p upper bound transfers through the diagonal isometry $\ell_p \hookrightarrow S_p$.

1 Source question

The source is Francisco Escudero Gutierrez and Garazi Muguruza, *All S_p notions of quantum expansion are equivalent*, arXiv:2405.03517v1 (2024), Question 2 in Section 3.2, PDF pages 4–5 [1]. It asks:

Let $p \in [1, \infty)$. What is the minimum distortion D_{n,S_p} needed, as a function of n and p , to embed any n -point metric space into S_p ?

The following are direct crops from the source PDF. They include the question, the definition of $\tilde{h}_{S_p}$, and the proposed expander route to a lower bound.

3.2 What about S_p ?

As pointed out by Li et al. [LQW⁺22], our Theorem 1 can be regarded as a quantum analogue of Eq. (6) proved by Matoušek, in the sense that both prove relations between notions of expansion based on similar norms. However, we believe that this resemblance is notational, rather than operational, and so our Theorem 1 does not seem to help to construct metric spaces that are hard to embed into S_p . To the best of our knowledge there is no literature about this problem, so we pose it as an open question.

Question 2. *Let $p \in [1, \infty)$. What is the minimum distortion D_{n, S_p} needed, as a function of n and p , to embed any n -point metric space into S_p ?*

To answer this question it would suffice to show that the following family of notions of graph expansion based on S_p norms for $p \in [1, \infty)$ are related in a certain way

$$\tilde{h}_{S_p}(G) := \min_{f: G \rightarrow S_p} \left(\frac{\frac{1}{|E|} \sum_{(i,j) \in E} \|f(i) - f(j)\|_{S_p}^p}{\frac{1}{n^2} \sum_{i,j \in [n]} \|f(i) - f(j)\|_{S_p}^p} \right)^{1/p}.$$

Indeed, as S_2 is isometric to ℓ_2 , we have that there exists a sequence of n -vertex d -regular graphs G_n such that $\inf \tilde{h}_{S_2}(G_n) > 0$. Hence, if one manages to show that for every d -regular graph

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G it holds that

$$\tilde{h}_{S_p}(G) = \Omega_d \left(\frac{\tilde{h}_{S_2}(G)}{p} \right),$$

we would have that $\inf \tilde{h}_{S_p}(G_n) = \Omega(1/p)$, from which we could reproduce the argument of Matoušek and prove that $D_{n, S_p} = \Omega(\log(n)/p)$. The matching upper bound follows from the fact that ℓ_p can be embedded isometrically into S_p .

2 Proof intuition

There are two complementary regimes. For $p \geq 2$, the mixed function space $L_p(V; S_p)$ is itself a noncommutative L_p space. Interpolating a centered lazy random walk between its Hilbert spectral gap and its L_∞ bound makes a k -step walk contractive once $k = O(p/\delta)$. Jensen's inequality and a path decomposition then turn this into the desired one-step Poincare inequality.

That interpolation necessarily degenerates at $p = 1$. For $1 \leq p \leq 2$, translate the map at a vertex chosen by averaging and apply the noncommutative Mazur map into the Hilbert space S_2 . Its inverse estimate contains a radial weight r^{2-p} . The chosen origin bounds the total radial moment, and a weighted Holder inequality shows that at least half of the original pairwise p -energy survives after the transformation. The ordinary Hilbert Poincare inequality then gives a bound uniform all the way to the trace-class endpoint.

3 Definitions and results

Let $G = (V, E)$ be a finite undirected d -regular graph, $n = |V|$, and let P be its normalized adjacency matrix. Write

$$\delta = 1 - \lambda_2(P),$$

where $\lambda_2(P)$ is the largest eigenvalue on the orthogonal complement of the constants. For $1 \leq p < \infty$, set

$$\tilde{h}_{S_p}(G) = \inf_{f: V \rightarrow S_p} \frac{\left(\frac{1}{|E|} \sum_{\{i,j\} \in E} \|f(i) - f(j)\|_p^p \right)^{1/p}}{\left(\frac{1}{n^2} \sum_{i,j \in V} \|f(i) - f(j)\|_p^p \right)^{1/p}},$$

where constant maps are omitted.

Theorem 1 (Schatten-valued graph Poincare inequalities). *There is a universal constant $K \geq 1$ such that every finite regular graph with $\delta > 0$ satisfies*

$$\tilde{h}_{S_p}(G) \geq \begin{cases} (\delta/K)^{1/p}, & 1 \leq p \leq 2, \\ \delta/(14p), & 2 \leq p < \infty. \end{cases}$$

In particular, on any fixed-gap expander family the first bound is uniform for $1 \leq p \leq 2$, and the second is $\Omega(1/p)$ for $p \geq 2$.

The source proposes the stronger graph-by-graph comparison $\tilde{h}_{S_p}(G) \gtrsim_d \tilde{h}_{S_2}(G)/p$. Since $\tilde{h}_{S_2}(G) = \sqrt{\delta}$, our bounds have weaker dependence on a gap tending to zero. They have the precise uniformity needed on fixed-gap expanders and hence suffice to answer the metric question.

Corollary 2 (Full answer to Question 2). *There are universal constants $c, C > 0$ such that for every $n \geq 2$ and every $1 \leq p < \infty$,*

$$c \max \left\{ 1, \frac{\log n}{p} \right\} \leq D_{n, S_p} \leq C \max \left\{ 1, \frac{\log n}{p} \right\}.$$

4 Mazur-map lemmas for the endpoint range

The lazy-walk interpolation proof for $p > 1$ degenerates as $p \downarrow 1$. The replacement is nonlinear. The Mazur map sends S_p to the Hilbert space S_2 , but for $p < 2$ its inverse Lipschitz estimate carries a radial weight. Choosing the translation point by averaging makes the total radial moment no larger than the pairwise energy. A second Holder inequality then absorbs the weight without any loss depending on p .

For the polar decomposition $x = u|x|$, define

$$M_{p,2}(x) = u|x|^{p/2}.$$

The identity $M_{2,p}(M_{p,2}(x)) = x$ will transfer quantitative control back from S_2 to S_p .

Lemma 3 (Uniform Mazur estimates). *There is a universal constant $A \geq 1$ such that, for every $1 \leq p \leq 2$ and every $x, y \in S_p$,*

$$\|M_{p,2}(x) - M_{p,2}(y)\|_2 \leq A \|x - y\|_p^{p/2}, \quad (1)$$

$$\|x - y\|_p \leq A \max\{\|x\|_p, \|y\|_p\}^{1-p/2} \|M_{p,2}(x) - M_{p,2}(y)\|_2. \quad (2)$$

The usual value $0^0 = 1$ may be used at $p = 2$.

Proof. Ricard proved that the noncommutative Mazur map $M_{r,s}$ is $\min\{r/s, 1\}$ -Holder on the unit ball of every noncommutative L_r -space [5]. The quantitative proof gives a constant depending only on the target exponent s when $r < s$, and an absolute constant times r/s when $r > s$; see in particular the commutator estimate in Lemma 2.6 of [5]. Thus the constants for $M_{p,2}$, $1 \leq p \leq 2$, and for $M_{2,p}$, $1 \leq p \leq 2$, are bounded by one universal constant.

The unit-ball estimate for $M_{p,2}$, together with its homogeneity of degree $p/2$, gives (1) for arbitrary x, y . For the inverse estimate, put $a = M_{p,2}(x)$, $b = M_{p,2}(y)$, and $R = \max\{\|a\|_2, \|b\|_2\}$. Homogeneity of $M_{2,p}$ and its unit-ball Lipschitz bound give

$$\|M_{2,p}(a) - M_{2,p}(b)\|_p \leq AR^{2/p-1}\|a - b\|_2.$$

Since $\|a\|_2 = \|x\|_p^{p/2}$ and $M_{2,p}(a) = x$, this is precisely (2). The zero case follows by continuity. At $p = 2$, both maps are the identity. $\square$

The next elementary estimate is the step that removes the radial weight.

Lemma 4 (Weighted pair averaging). *Let $1 \leq p < 2$, $q = 2 - p$, and let $x_i \in S_p$ for $i \in V$. Write*

$$d_{ij} = \|x_i - x_j\|_p, \quad r_i = \|x_i\|_p, \quad B = \mathbb{E}_{i,j} d_{ij}^p.$$

If $\mathbb{E}_i r_i^p \leq B$, then, with a zero summand when its denominator vanishes,

$$\mathbb{E}_{i,j} \frac{d_{ij}^2}{r_i^q + r_j^q} \geq \frac{B}{2}.$$

Proof. Put $s_{ij} = r_i^q + r_j^q$ and $C = \mathbb{E}_{i,j} d_{ij}^2 / s_{ij}$. Holder's inequality with conjugate exponents $2/p$ and $2/q$ gives

$$\begin{aligned} B &= \mathbb{E}_{i,j} \left(\frac{d_{ij}^2}{s_{ij}} \right)^{p/2} s_{ij}^{p/2} \\ &\leq C^{p/2} \left(\mathbb{E}_{i,j} s_{ij}^{p/q} \right)^{q/2}. \end{aligned}$$

Because $p/q \geq 1$,

$$(r_i^q + r_j^q)^{p/q} \leq 2^{p/q-1} (r_i^p + r_j^p).$$

Consequently

$$\mathbb{E}_{i,j} s_{ij}^{p/q} \leq 2^{p/q} \mathbb{E}_i r_i^p \leq 2^{p/q} B.$$

It follows that

$$B \leq 2^{p/2} C^{p/2} B^{q/2}.$$

If $B > 0$, division and raising to the power $2/p$ give $B \leq 2C$. The case $B = 0$ is immediate. $\square$

5 Proof of Theorem 1 for $1 \leq p \leq 2$

Fix $f : V \rightarrow S_p$, and abbreviate

$$d_{ij} = \|f(i) - f(j)\|_p, \quad B = \mathbb{E}_{i,j} d_{ij}^p.$$

There is a vertex $o \in V$ such that

$$\mathbb{E}_i d_{io}^p \leq B,$$

because the average of the left side over o equals B . Translate at this vertex: $x_i = f(i) - f(o)$, $r_i = \|x_i\|_p = d_{io}$, and set

$$a_i = M_{p,2}(x_i), \quad h_{ij} = \|a_i - a_j\|_2.$$

Suppose first that $p < 2$, and put $q = 2 - p$. Squaring (2) gives

$$h_{ij}^2 \geq A^{-2} \frac{d_{ij}^2}{\max\{r_i, r_j\}^q} \geq A^{-2} \frac{d_{ij}^2}{r_i^q + r_j^q}.$$

Lemma 4 therefore implies

$$\mathbb{E}_{i,j} h_{ij}^2 \geq \frac{B}{2A^2}. \quad (3)$$

On the other hand, (1) gives on every edge

$$h_{ij}^2 \leq A^2 d_{ij}^p. \quad (4)$$

The Hilbert-space Poincare inequality for the normalized adjacency matrix is

$$\mathbb{E}_{i,j} \|a_i - a_j\|_2^2 \leq \frac{1}{\delta} \frac{1}{|E|} \sum_{\{i,j\} \in E} \|a_i - a_j\|_2^2. \quad (5)$$

Indeed, both sides equal twice the corresponding variance and Dirichlet form normalization. Combining (3), (4), and (5) yields

$$B \leq \frac{2A^4}{\delta} \frac{1}{|E|} \sum_{\{i,j\} \in E} d_{ij}^p.$$

Thus the first line of Theorem 1 holds with $K = 2A^4$. For $p = 2$, it is just (5), so increasing K if necessary covers the closed interval $1 \leq p \leq 2$. Notice that no limiting argument is used at $p = 1$.

6 Proof of Theorem 1 for $p \geq 2$

For completeness, we include the independent interpolation argument that has the correct $1/p$ dependence in the high- p range. Put $Q = (I + P)/2$, and let J be averaging over V . All eigenvalues of Q lie in $[0, 1]$, and on the orthogonal complement of constants,

$$\alpha := \|Q - J\|_{\ell_2(V) \rightarrow \ell_2(V)} = 1 - \frac{\delta}{2} < 1.$$

Moreover, $QJ = JQ = J$, so $Q^k - J = (Q - J)^k$.

Consider the semifinite von Neumann algebra

$$\mathcal{A} = \ell_\infty(V) \overline{\otimes} B(\ell_2)$$

with normalized trace in the V -coordinate. Then $L_r(\mathcal{A}) = L_r(V; S_r)$ isometrically. The maps Q^k and J are contractions on $L_\infty(\mathcal{A})$, whereas at the Hilbert endpoint

$$\|Q^k - J\|_{L_2(\mathcal{A}) \rightarrow L_2(\mathcal{A})} = \alpha^k.$$

Noncommutative complex interpolation [3] gives, for $p \geq 2$,

$$\|Q^k - J\|_{L_p(\mathcal{A}) \rightarrow L_p(\mathcal{A})} \leq 2^{1-2/p} \alpha^{2k/p}. \quad (6)$$

Let $L = -\log \alpha$, interpreting $\alpha = 0$ by taking $k = 1$, and choose

$$k = \max \left\{ 1, \left\lceil \frac{(p-1) \log 2}{L} \right\rceil \right\}.$$

Then (6) is at most $1/2$. For $f : V \rightarrow S_p$, set $\bar{f} = Jf$ and $z = f - \bar{f}$. Centering, the contraction, Jensen's inequality, and Minkowski's inequality give

$$\begin{aligned} (\mathbb{E}_{i,j} \|f(i) - f(j)\|_p^p)^{1/p} &\leq 2 \|z\|_{L_p(V; S_p)} \\ &\leq 4 \left(\frac{1}{n} \sum_{i,j} (Q^k)_{ij} \|f(i) - f(j)\|_p^p \right)^{1/p}. \end{aligned} \quad (7)$$

Let $(X_t)_{t=0}^k$ be the stationary lazy random walk with transition matrix Q . The triangle inequality and convexity imply

$$\|f(X_0) - f(X_k)\|_p^p \leq k^{p-1} \sum_{t=1}^k \|f(X_{t-1}) - f(X_t)\|_p^p.$$

After averaging, stationarity shows that the k -step energy in (7) is at most k^p times the one-step lazy energy. Its non-loop part is one half of the source edge energy. Dropping this harmless factor gives

$$(\mathbb{E}_{i,j} \|f(i) - f(j)\|_p^p)^{1/p} \leq 4k \left(\frac{1}{|E|} \sum_{\{i,j\} \in E} \|f(i) - f(j)\|_p^p \right)^{1/p}.$$

Thus $\tilde{h}_{S_p}(G) \geq 1/(4k)$. Finally, $-\log(1 - \delta/2) \geq \delta/2$ and $0 < \delta \leq 2$, so

$$k \leq \frac{(2 + 2 \log 2)p}{\delta}.$$

Since $4(2 + 2 \log 2) < 14$, this proves the second line of Theorem 1.

7 Proof of Corollary 2

Fix a constant-degree family of classical expanders with spectral gaps at least $\delta_0 > 0$ [2]. In a d -regular n -vertex graph, a ball of radius r contains at most $1 + d + \dots + d^r$. Hence a fixed positive fraction of ordered vertex pairs have graph distance $\Omega_d(\log n)$, and their p -root mean distance is $\Omega_d(\log n)$, uniformly in $p \geq 1$.

Suppose that the graph metric embeds in S_p with distortion D , and rescale the embedding to be nonexpanding. The edge-energy root is at most 1, while the all-pairs energy root is at least $c_d \log n / D$. When $1 \leq p \leq 2$, Theorem 1 bounds the latter by a universal constant because $(\delta_0/K)^{1/p} \geq \delta_0/K$. Thus

$$D \gtrsim \log n \quad (1 \leq p \leq 2).$$

When $p \geq 2$, the high- p graph bound gives

$$D \gtrsim \frac{\log n}{p}.$$

Together with $D \geq 1$, these estimates are $D \gtrsim \max\{1, \log n/p\}$ throughout the full range. Standard fixed-degree expanders at all sufficiently large admissible sizes, and the monotonicity of worst-case distortion, give the conclusion for every n .

For the upper bound, Matousek’s extension of Bourgain’s construction gives distortion $O(\log n/p)$ into ℓ_p in the range $p \lesssim \log n$ [4]; the source paper recalls the same estimate [1]. The diagonal map $\ell_p \hookrightarrow S_p$ is isometric. When $p \geq \log n$, the Frechet embedding into ℓ_∞^n , followed by the identity into ℓ_p^n , has distortion at most $n^{1/p} \leq e$. Enlarging the universal constant handles the finitely many small cases and proves the matching upper bound.

8 Verification, limitations, and novelty

The weighted averaging computation and finite matrix instances of the Mazur-energy argument were checked by `code/check_constants.py`; these checks are consistency tests, not proof. A separate normalization and endpoint audit appears in `verification.md`. The original source PDF and Ricard’s supporting paper are included beside this packet.

The conclusion is full relative to Question 2: it determines the sharp joint asymptotics in n and p for every $1 \leq p < \infty$. It does not prove the stronger graph-by-graph comparison suggested in the source; for a small spectral gap our displayed dependence on δ is weaker.

A bounded novelty search was performed on 19 July 2026. The run indexes were searched using arXiv:2405.03517, its exact title, D_{n,S_p} , Schatten distortion, noncommutative Mazur maps, trace norm, and graph Poincare keywords. Targeted arXiv/web searches used the exact source question and close phrases. Ricard’s quantitative Mazur-map paper (arXiv:1407.8334), Mendel–Naor’s superexpander paper (arXiv:1207.4705), and related Schatten nonembedding results were checked. No later paper explicitly settling Question 2, including its S_1 endpoint, was found. Novelty confidence is moderate because this was a bounded search rather than an exhaustive citation review.

The most important human-review point is the uniformity in $p \in [1, 2]$ of Lemma 3. It follows from the constants in Ricard’s proof, not merely from the qualitative statement of his theorem. The remaining new ingredient, Lemma 4, is proved above with the explicit constant $1/2$.

References

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